

RAGHAV KACHROO

(858)-241-1760 | rkachroo@ucsd.edu | linkedin.com/in/raghavkachroo | github.com/mister-raggs

Software engineer drawn to making ML systems fast in production - coming from distributed systems and data pipelines, working down the stack toward inference optimization, compilers, and GPUs.

EXPERIENCE

Hao AI Lab, UC San Diego

Jan 2026 – Present

Contributor

San Diego, CA

FastVideo — open-source video-generation inference and post-training framework; **4k+** stars, supported as an NVIDIA Dynamo custom worker and forked into SGLang's multimodal-generation stack. [Link to PRs](#)

- Added opt-in **cache-dit** step caching to Wan DiT, cutting wall time **18-32%** at SSIM **0.94-0.96** on Wan2.1-1.3B.
- Registered **torch.library** custom ops with **register_autograd** across FA2's default and masked/varlen paths, making attention traceable under **torch.compile** for inference and training.
- Quantized decoded frames to uint8 on-device before the GPU-to-CPU transfer, cutting end-to-end video-inference latency **15.6%** on Cosmos 2.5 2B/A100 at MS-SSIM 1.0 across all frames.
- Built a A/B harness and documentation for **torch.compile** path, surfacing **23.7%** lower latency on Wan2.1-1.3B.
- Validated FastVideo on NVIDIA DGX Spark and upstreamed a bf16 VAE-decode path at **1.2-1.3x** decode.

Amazon

Jun 2025 – Sep 2025

Software Development Engineer Intern

Bellevue, WA

- Built a distributed log indexing and query service over **42M+** log entries/hour, executing binary-search reads concurrently across a worker pool to reduce incident triage latency from **15+ minutes to under 45 seconds**.
- Automated on-call SOPs using AWS Step Functions and Lambda, saving **12+ engineer-hours per week**.
- Exposed the log query service via MCP with caching and query batching, maintaining **sub-2s** observed response times during concurrent incident investigations.

Aark Global

Apr 2023 – Sep 2024

Software Developer, AI/ML

Delhi, India

- Built an async document ingestion pipeline processing **18,000+ pages/day** via Azure Queue Storage worker pool, maintaining throughput under variable load.
- Implemented a phased Cosmos DB-to-MongoDB migration using dual writes and historical backfill while serving reads from Cosmos DB, maintaining **sub-100ms P95** latency through cutover.
- Built a full-text search pipeline ingesting scanned PDFs through OCR into indexed Elasticsearch documents, enabling **sub-180ms** query latency over previously unsearchable content.

EDUCATION

University of California, San Diego

Sep 2024 – Mar 2026

Master of Science in Data Science

Indraprastha Institute of Information Technology, Delhi

Sep 2022 – Sep 2023

Post Graduate Diploma in Data Science & Artificial Intelligence

University of Delhi

Jul 2018 – Jun 2021

Bachelor of Management Studies

PROJECTS & RESEARCH

Flare: ML-Powered Anomaly Detection Pipeline | [GitHub](#)

Jan 2026 – June 2026

- Built an end-to-end ML anomaly detection pipeline applying Drain3 template mining, Isolation Forest scoring, and DBSCAN clustering to **11.2M log lines** across **575K blocks**, achieving **0.642 F1** on a held-out split at **~40ms** scoring latency per block (CPU-only).

OneBusAway / maglev: OSS Contributions | [GitHub](#)

Jul 2026 – Present

- Landed **9** upstream fixes to this Go transit REST API from **9** spec-conformance issues I filed - query radius, sort order, **maxCount** clamping, route-type filtering, and **includeReferences** handling.

TRAJ: AI Mobile App to Reduce the Gap Between Triage and Care | [IEEE Publication](#)

Jul 2025

- Co-authored a IEEE paper on an LLM-based mobile application for patient triage and clinical data gathering.

TECHNICAL SKILLS

Languages: Python, Go, SQL, Java

ML Systems: PyTorch, torch.library, FlashAttention, Nsight Systems, Diffusers, Quantization

Backend & Infra: FastAPI, REST APIs, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, MCP

Distributed & Data Systems: Kafka, Spark, Airflow, Elasticsearch, MongoDB, Cosmos DB, Azure Queue Storage, Redis

Observability & Cloud: Prometheus, OpenTelemetry, AWS (Lambda, Step Functions, S3), Azure

Concepts: Distributed Systems, Concurrency, Inference Optimization, MLOps, System Design, RAG